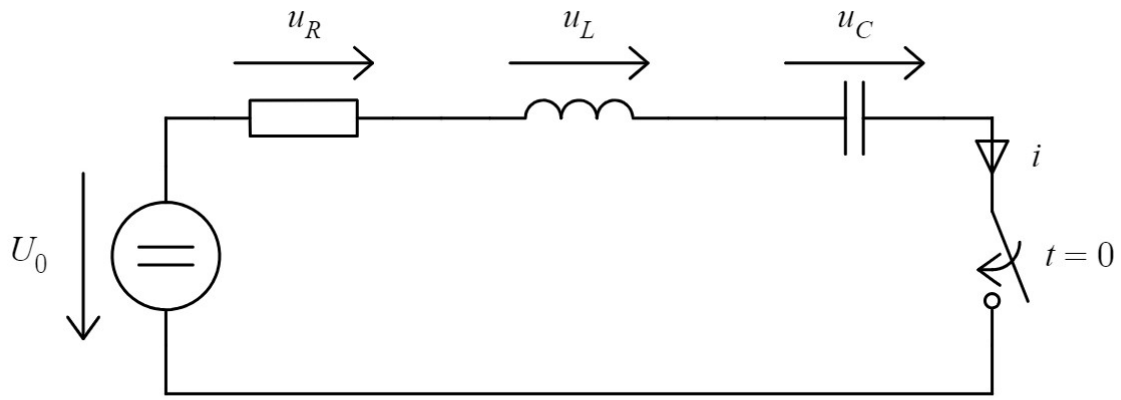



Z II. Kirchhoffova zákona lze napsat rovnice:

$$u_R(t) + u_L(t) + u_C(t) = U_0$$
$$R \cdot i(t) + L \cdot \frac{di(t)}{dt} + \frac{1}{C} \cdot \int_0^t i(x)\,dx = U_0$$

Derivováním a přerovnáním této rovnice dostaneme rovnici tuto:

$$L \cdot \frac{d^2 i(t)}{dt^2} + R \cdot \frac{di(t)}{dt} + \frac{1}{C} \cdot i(t) = 0$$

Řešení diferenciální rovnice bude ve tvaru: $i(t) = k \cdot e^{\lambda t}$, kde $\lambda$ uspokojuje tuto kvadratickou rovnici:

$$L \cdot \lambda^2 + R \cdot \lambda + \frac{1}{C} = 0$$

$$\lambda_{1,2} = \frac{-R \pm \sqrt{R^2 - \frac{4L}{C}}}{2L} = \frac{-R}{2L} \pm \sqrt{\frac{R^2}{4L^2} - \frac{1}{LC}}$$
$$\lambda_1 = \frac{-R}{2L} + \sqrt{\frac{R^2}{4L^2} - \frac{1}{LC}}$$
$$\lambda_2 = \frac{-R}{2L} - \sqrt{\frac{R^2}{4L^2} - \frac{1}{LC}}$$

Dosazením: $\alpha = \frac{R}{2L}, \omega_0 = \frac{1}{\sqrt{LC}}, \beta = \sqrt{\alpha^2 - \omega_0^2},$ získáme:

$$\lambda_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2} = -\alpha \pm \beta$$

$$i(t) = k_1 e^{t\lambda_1} + k_2 e^{t\lambda_2} = k_1 e^{t(-\alpha+\beta)} + k_2 e^{t(-\alpha-\beta)}$$

Vytknutím poté získáme:

$$i(t) = e^{-\alpha t}(k_1 e^{+t\beta} + k_2 e^{-t\beta})$$

Koeficienty $k_1$ a $k_2$ zjistíme ze spojitosti průběhu proudu cívkou a z počáteční podmínky v první rovnici a dostáváme: $i(0) = 0$ a pro $t = 0$ :

$$\begin{aligned} R \cdot i(0) + L \cdot \frac{di(t)}{dt}\Big|_{t=0} + \frac{1}{C} \cdot \int_0^0 i(x)\,dx &= U_0 \\ R \cdot 0 + L \cdot \frac{di(t)}{dt}\Big|_{t=0} + \frac{1}{C} \cdot 0 &= U_0 \\ L \cdot \frac{di(t)}{dt}\Big|_{t=0} &= U_0 \\ \frac{di(t)}{dt}\Big|_{t=0} &= \frac{U_0}{L} \end{aligned}$$

Máme tedy soustavu rovnic:

$$\begin{cases} k_1 \cdot e^0 + k_2 \cdot e^0 = 0 \\ k_1\lambda_1 \cdot e^0 + k_2\lambda_2 \cdot e^0 = \dfrac{U_0}{L} \end{cases}$$

Vyřešením soustavy rovnic dostaneme:

$$\begin{aligned} k_1 &= \frac{U_0}{2\beta L} \\ k_2 &= \frac{-U_0}{2\beta L} \end{aligned}$$

Užitím $\beta = \sqrt{\alpha^2 - \omega_0^2}$

$$i(t) = \frac{U_0}{2\beta L} e^{-\alpha t}(e^{+t\beta} - e^{-t\beta})$$

Podle znaménka $\frac{R^2}{4L^2} - \frac{1}{LC}$ určíme chování obvodu:

$$\begin{aligned} \frac{R^2}{4L^2} - \frac{1}{LC} &= 0 \\ \frac{R^2}{4L^2} &= \frac{1}{LC} \\ R^2 &= \frac{4L}{C} \\ R_{mez} &= 2\sqrt{\frac{L}{C}} \end{aligned}$$

1) Přetlumený obvod

$R > R_{mez}$

$\lambda \in \mathbb{R}$

$$\begin{aligned} i(t) &= \frac{U_0}{2\beta L} e^{-\alpha t}(e^{+t\beta} - e^{-t\beta}) \\ &= \frac{U_0}{\beta L} e^{-\alpha t} \sinh(\beta t) \end{aligned}$$

$$\begin{aligned} u_R(t) &= \frac{RU_0}{2\beta L} e^{-\alpha t}(e^{+t\beta} - e^{-t\beta}) \\ &= \frac{RU_0}{\beta L} e^{-\alpha t} \sinh(\beta t) \end{aligned}$$

$$\begin{aligned} u_L(t) &= \frac{U_0}{2\beta} e^{-\alpha t}\left(\beta(e^{+\beta t} + e^{-\beta t}) - \alpha(e^{+\beta t} - e^{-\beta t})\right) \\ &= \frac{U_0}{\beta} e^{-\alpha t}(\beta \cosh(\beta t) - \alpha \sinh(\beta t)) \end{aligned}$$

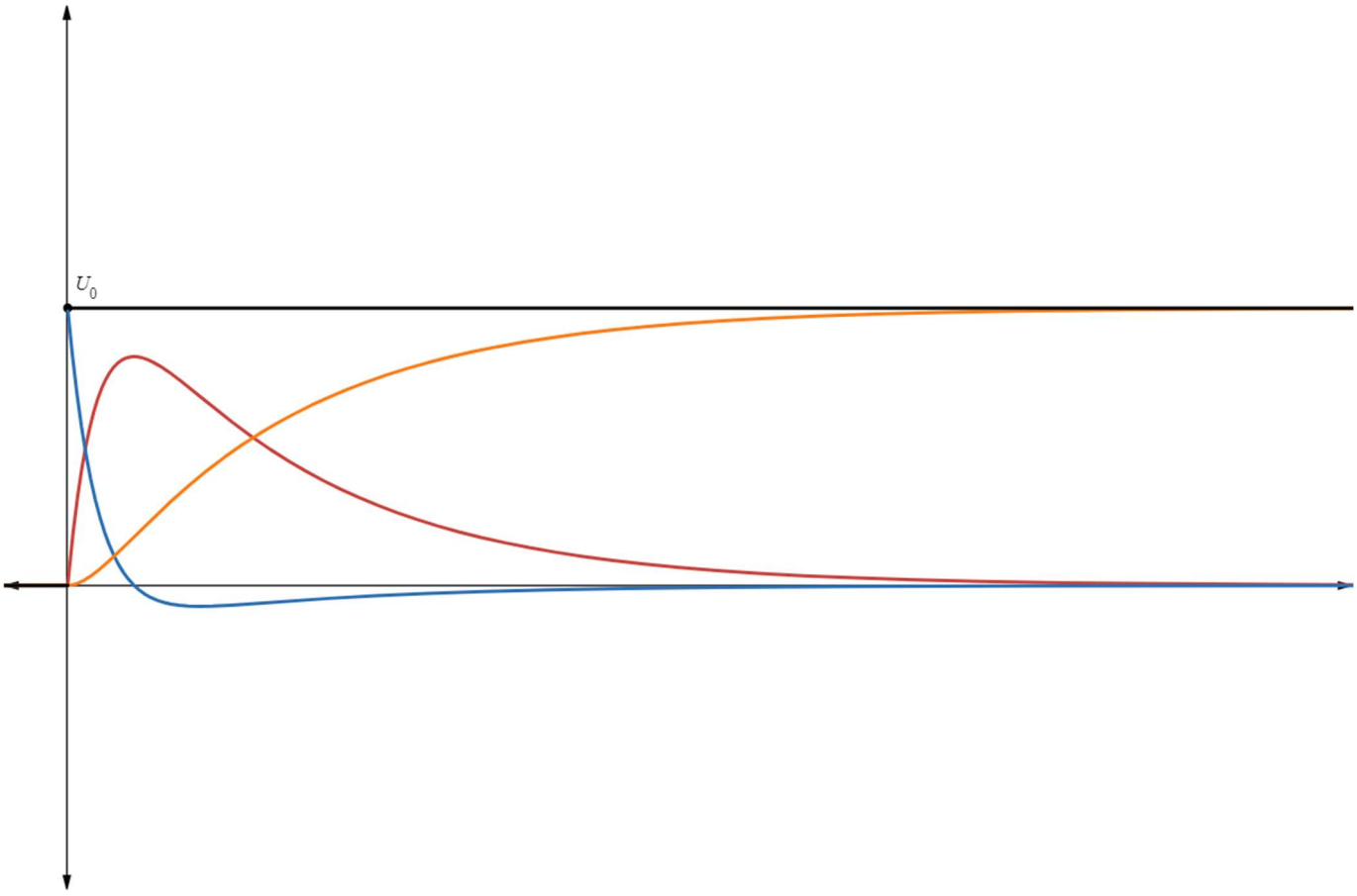
$$\begin{aligned} u_C(t) &= \frac{U_0}{2\beta LC} e^{-\alpha t}\left(\frac{e^{\alpha t} - e^{+\beta t}}{\alpha - \beta} - \frac{e^{\alpha t} - e^{-\beta t}}{\alpha + \beta}\right) \\ &= \frac{U_0}{\beta LC(\beta^2 - \alpha^2)} e^{-\alpha t}(\beta \cosh(\beta t) + \alpha \sinh(\beta t) - \beta e^{+\alpha t}) \end{aligned}$$

$u_R(x)$

$u_L(x)$

$u_C(x)$

$u_R(x)+u_L(x)+u_C(x)$



2) Kriticky tlumený obvod

$R = R_{mez}$

$\lambda = -\alpha, \beta = 0$

$$i(t) = \lim_{\beta\to 0}\left(\frac{U_0}{2\beta L}e^{-\alpha t}(e^{+t\beta} - e^{-t\beta})\right) = \frac{U_0}{2L}e^{-\alpha t}\lim_{\beta\to 0}\left(\frac{e^{+t\beta} - e^{-t\beta}}{\beta}\right)$$

$$\overset{L'H}{=} \frac{U_0}{2L}e^{-\alpha t}\lim_{\beta\to 0}\left(\frac{+te^{+t\beta} - (-t)e^{-t\beta}}{1}\right) = \frac{U_0}{2L}e^{-\alpha t}\lim_{\beta\to 0}\left(\frac{t\cdot 1 + t\cdot 1}{1}\right) = \frac{U_0}{2L}e^{-\alpha t}\cdot 2t$$

$$i(t) = \frac{U_0}{L}te^{-\alpha t}$$

$$u_R(t) = \frac{RU_0}{L}te^{-\alpha t}$$

$$u_L(t) = U_0(1-\alpha t)e^{-\alpha t}$$

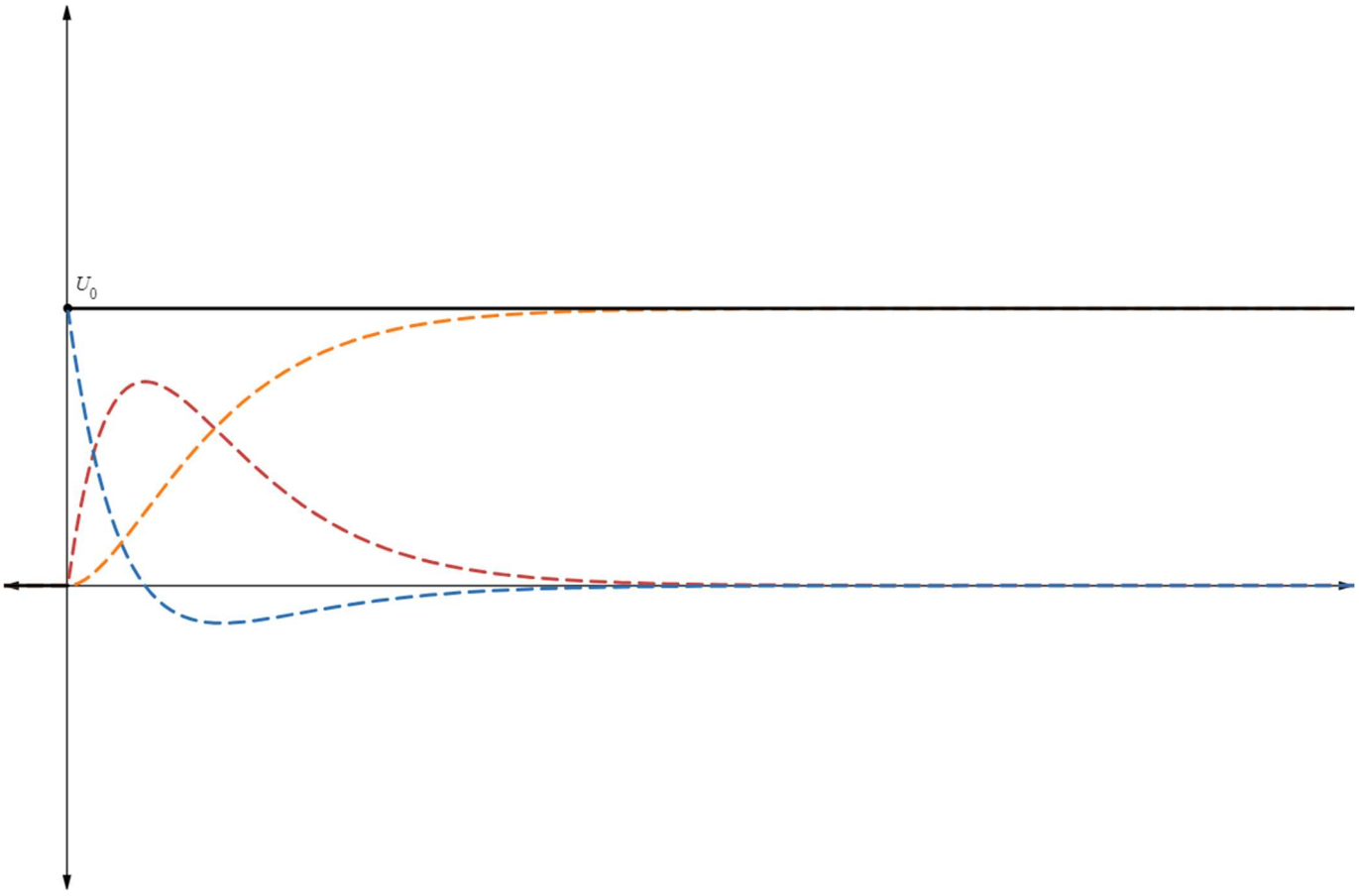
$$u_C(t) = \frac{U_0}{\alpha^2 CL}(1-(1+\alpha t)e^{-\alpha t})$$

$u_R(x)$

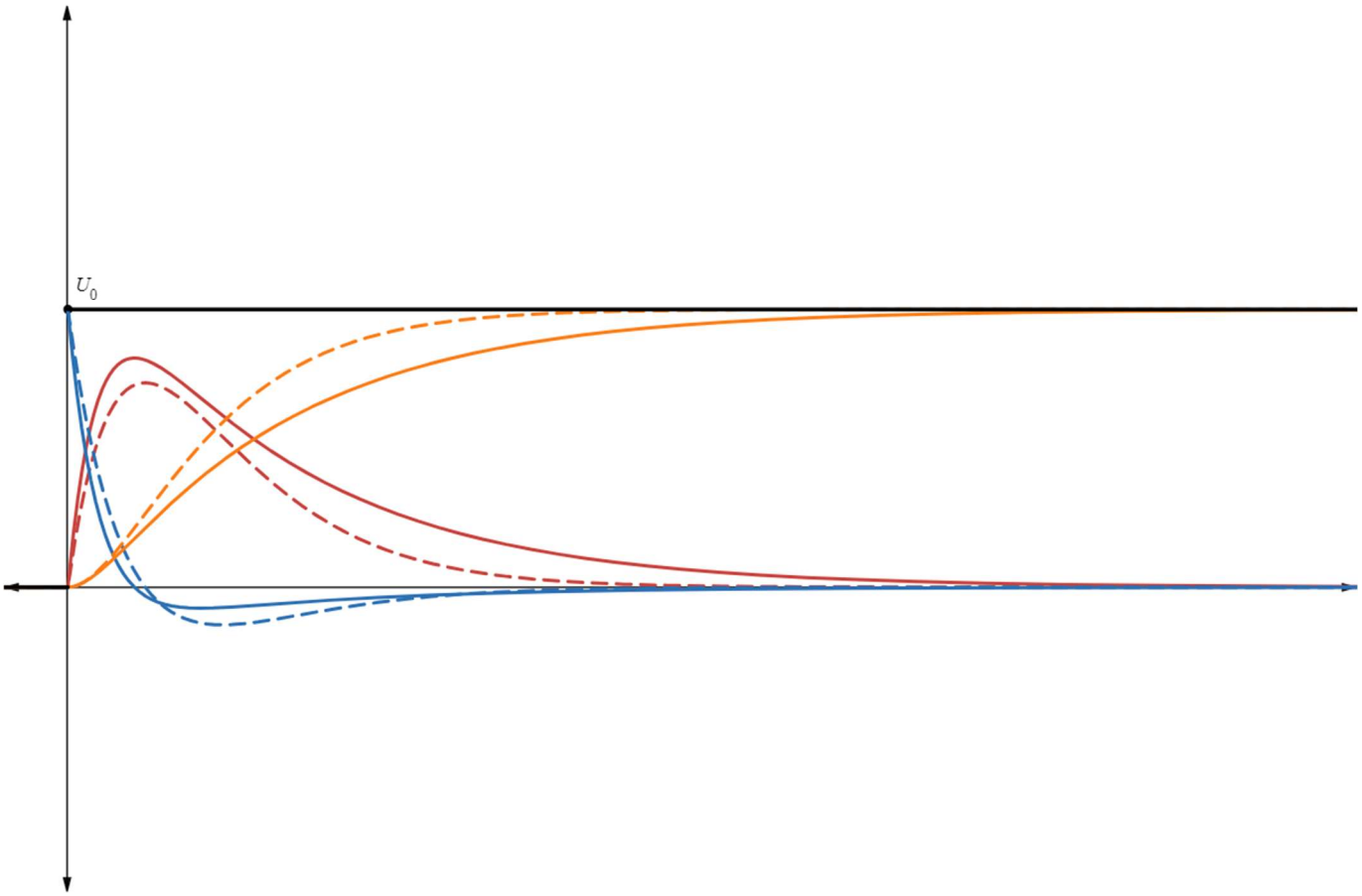
$u_L(x)$

$u_C(x)$

$u_R(x)+u_L(x)+u_C(x)$



Porovnání 1) a 2)



3) Podtlumený obvod

$R < R_{mez}$

$\lambda \in \mathbb{C} \quad a \quad \beta = j\omega$

$$\begin{aligned} i(t) &= \frac{U_0}{2j\omega L} e^{-\alpha t}(e^{+jt\omega} - e^{-jt\omega}) \\ &= \frac{U_0}{2j\omega L} e^{-\alpha t}(\cos(\omega t) + j\sin(\omega t) - \cos(-\omega t) - j\sin(-\omega t)) \\ &= \frac{U_0}{2j\omega L} e^{-\alpha t}(2j\sin(\omega t)) \\ &= \frac{U_0}{\omega L} e^{-\alpha t}\sin(\omega t) \end{aligned}$$

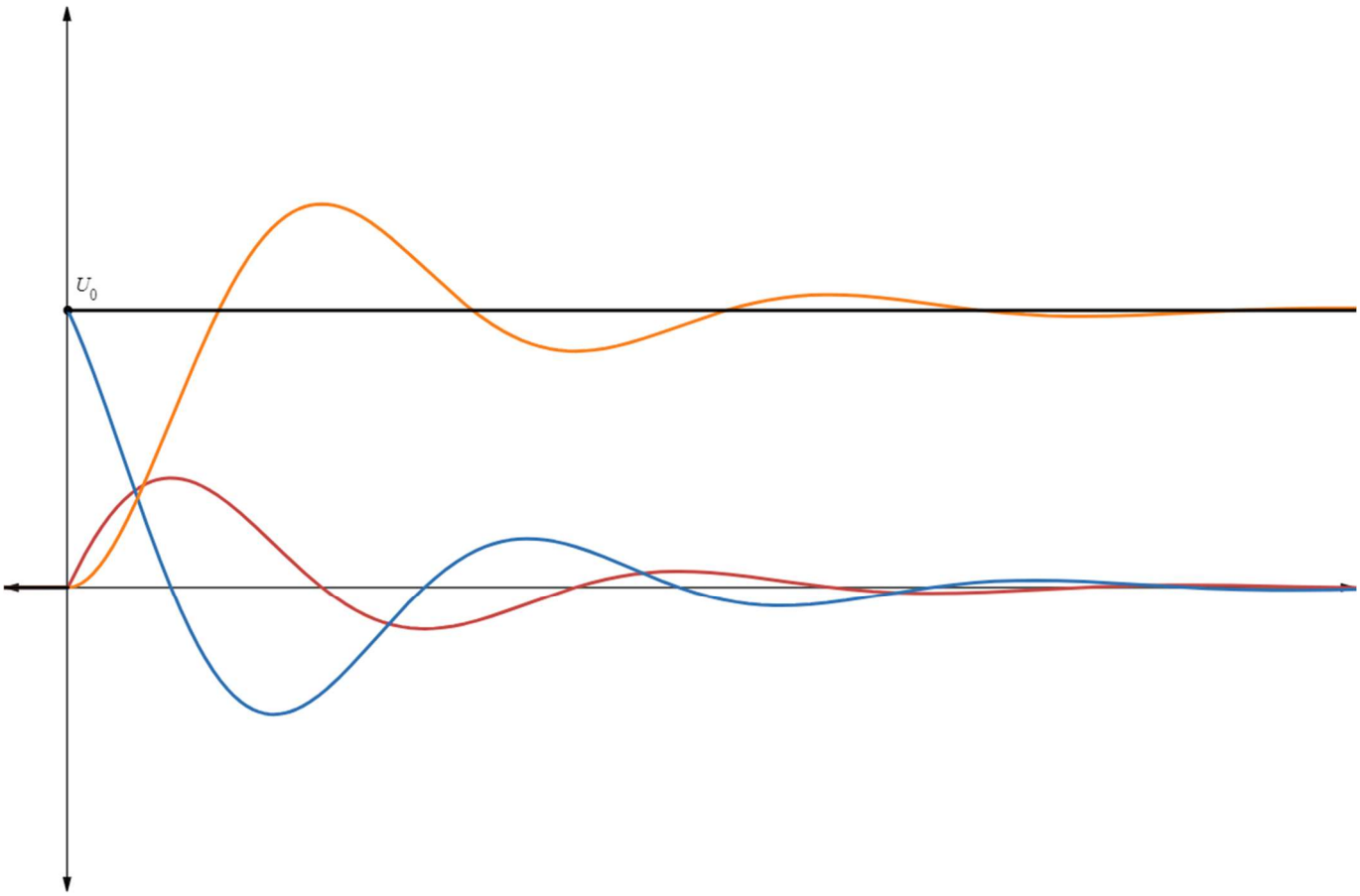
$$i(t) = \frac{U_0}{\omega L} e^{-\alpha t}\sin(\omega t)$$

$$u_R(t) = \frac{RU_0}{\omega L} e^{-\alpha t}\sin(\omega t)$$

$$u_L(t) = \frac{U_0}{\omega} e^{-\alpha t}(\omega\cos(\omega t) - \alpha\sin(\omega t))$$

$$u_C(t) = \frac{U_0}{\omega LC(\alpha^2 + \omega^2)} e^{-\alpha t}(\omega e^{+\alpha t} - \omega\cos(\omega t) - \alpha\sin(\omega t))$$



4) Netlumený obvod

$R = 0$

$\lambda \in \mathbb{C}, \alpha = 0 \quad a \quad \beta = j\omega_0$

$$i(t) = \frac{U_0}{\omega L} e^{-\alpha t} \sin(\omega t)$$
$$= \frac{U_0}{\omega_0 L} \sin(\omega_0 t)$$

$$i(t) = \frac{U_0}{\omega_0 L} \sin(\omega_0 t)$$

$$u_R(t) = \frac{RU_0}{\omega_0 L} \sin(\omega_0 t)$$

$$u_L(t) = U_0 \cos(\omega_0 t)$$

$$u_C(t) = U_0(1 - \cos(\omega_0 t))$$

